# Deep Learning

## Assignment # 2

SUBMITTED TO: Ma’am Nabeela

## ACLS CIFAR-10 Reproduction




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# Abstract

This project presents a reproduction study of selected experiments from the research paper “ACLS: Adaptive and Conditional Label Smoothing for Network Calibration” (ICCV 2023). The purpose is to examine the implementation and evaluation of a standard Cross-Entropy (CE) baseline and an ACLS-based model on the CIFAR-10 image-classification dataset. A ResNet-50 architecture is adapted to accept small 32 × 32 images and to produce ten class scores. The models are evaluated using classification accuracy, Expected Calibration Error (ECE), and Adaptive Expected Calibration Error (AECE), with reliability diagrams used to inspect confidence calibration.

The reproduction was conducted in a Kaggle notebook using PyTorch. The implementation uses a training/validation split from the CIFAR-10 training set and a held-out test set for final evaluation. The ACLS experiment required optimization-stability adjustments, including a reduced learning rate and gradient clipping. These changes mean that the experiment should be treated as a constrained reproduction rather than an exact replication of the original paper..


# 1. Introduction

Deep-learning classifiers can make correct predictions while assigning confidence values that do not reflect their actual likelihood of being correct. This is known as a calibration problem. For example, a model may frequently assign very high confidence to predictions that are incorrect. Calibration is important when model confidence is used to support decisions or to communicate uncertainty.

The selected paper proposes Adaptive and Conditional Label Smoothing (ACLS), a method intended to improve the calibration of neural-network predictions. This assignment implements a comparison between a conventional Cross-Entropy loss and an ACLS loss on CIFAR-10 using ResNet-50.

## 1.1 Objectives

- Prepare the CIFAR-10 dataset for image classification.
- Implement a ResNet-50 baseline trained with Cross-Entropy loss.
- Implement an ACLS-based training objective and run the experiment.
- Evaluate both models using Accuracy, ECE, and AECE.
- Compare the reproduced results with the reference values reported in the paper.
- Document implementation choices, limitations, and reproducibility challenges.

# 2. Dataset and Preprocessing

CIFAR-10 is a standard image-classification dataset containing 60,000 colour images across ten classes. Each image has a resolution of 32 × 32 pixels. The dataset provides 50,000 training images and 10,000 test images.

| Dataset property | Description |
|---|---|
| Dataset | CIFAR-10 |
| Total images | 60,000 |
| Training images | 50,000 |
| Test images | 10,000 |
| Image dimensions | 32 × 32 pixels, RGB |
| Classes | 10 |

A validation subset was separated from the training portion using a 10% validation fraction. The training pipeline applies random cropping with padding and random horizontal flipping. Images are converted to tensors and normalized using the CIFAR-10 channel statistics (mean: 0.4914, 0.4822, 0.4465; standard deviation: 0.2470, 0.2435, 0.2616). Evaluation data is normalized without random augmentation so that evaluation remains consistent.

The dataset was loaded through torchvision's CIFAR-10 utility. The notebook records the dataset preparation and transformation steps so the experiment can be repeated.

# 3. Model Architecture

ResNet-50 was selected as the classification architecture. It is a convolutional neural network that uses residual connections to help information and gradients pass through a deep network. The original ResNet-50 classifier is designed for a larger number of classes and larger input images, so it was adapted for CIFAR-10.

- The first convolution was changed to a 3 × 3 convolution with stride 1 and padding 1.
- The initial max-pooling operation was replaced with an identity operation.
- The final fully connected layer was changed to output ten class scores.

The same adapted architecture was used for the baseline and ACLS experiments so that the main comparison concerns the training loss rather than a change in network structure.

# 4. Methodology and Experimental Setup

## 4.1 Baseline: Cross-Entropy

The baseline model uses the standard Cross-Entropy loss. This loss compares the model's predicted class scores with the correct class labels and penalizes incorrect predictions. It provides the reference point for evaluating the ACLS implementation.

## 4.2 ACLS objective

The ACLS implementation combines the Cross-Entropy classification loss with additional regularization terms based on relationships between the predicted-class logit and other class logits. The regularization is controlled by the margin and the two weighting parameters. In this experiment, the configured values were margin = 6.0, $\lambda 1 = 0.1$, and $\lambda 2 = 0.01$. These values and the implemented loss calculation are recorded in the notebook.

## 4.3 Training configuration

| Parameter | Experiment setting |
|---|---|
| Dataset | CIFAR-10 |
| Network | ResNet-50 adapted for 10 classes |
| Optimizer | Stochastic Gradient Descent (SGD) |
| Batch size | 128 |
| Configured epochs | 5 |
| Momentum | 0.9 |
| Weight decay | 0.0005 |
| Validation fraction | 10% of training data |
| ACLS margin | 6.0 |
| ACLS λ1 | 0.1 |

| | |
|---|---|
| ACLS λ2 | 0.01 |
| Environment | Kaggle notebook; record actual GPU/CPU in experiment_config.json |
| ACLS learning rate | 0.0001 |
| Gradient clipping maximum norm | 1.0 |

The ACLS training initially showed numerical instability, including very large gradient values and non-finite loss. To make the experiment proceed, the learning rate was reduced and gradient clipping was introduced. The actual learning rate and other run-specific values should be taken from experiment_config.json because the stability-adjusted ACLS run did not use the same optimization conditions as the original paper.

### 4.4 Evaluation metrics

Accuracy is the percentage of test images classified correctly. ECE measures the weighted difference between average confidence and accuracy across equal-width confidence bins. AECE uses equal-count confidence bins, allowing calibration error to be assessed when bins contain similar numbers of predictions. The notebook uses 15 bins for these calculations.

A reliability diagram plots confidence against accuracy. A perfectly calibrated model would lie close to the diagonal reference line. Points below the diagonal indicate overconfidence in those confidence ranges, while points above it indicate underconfidence.

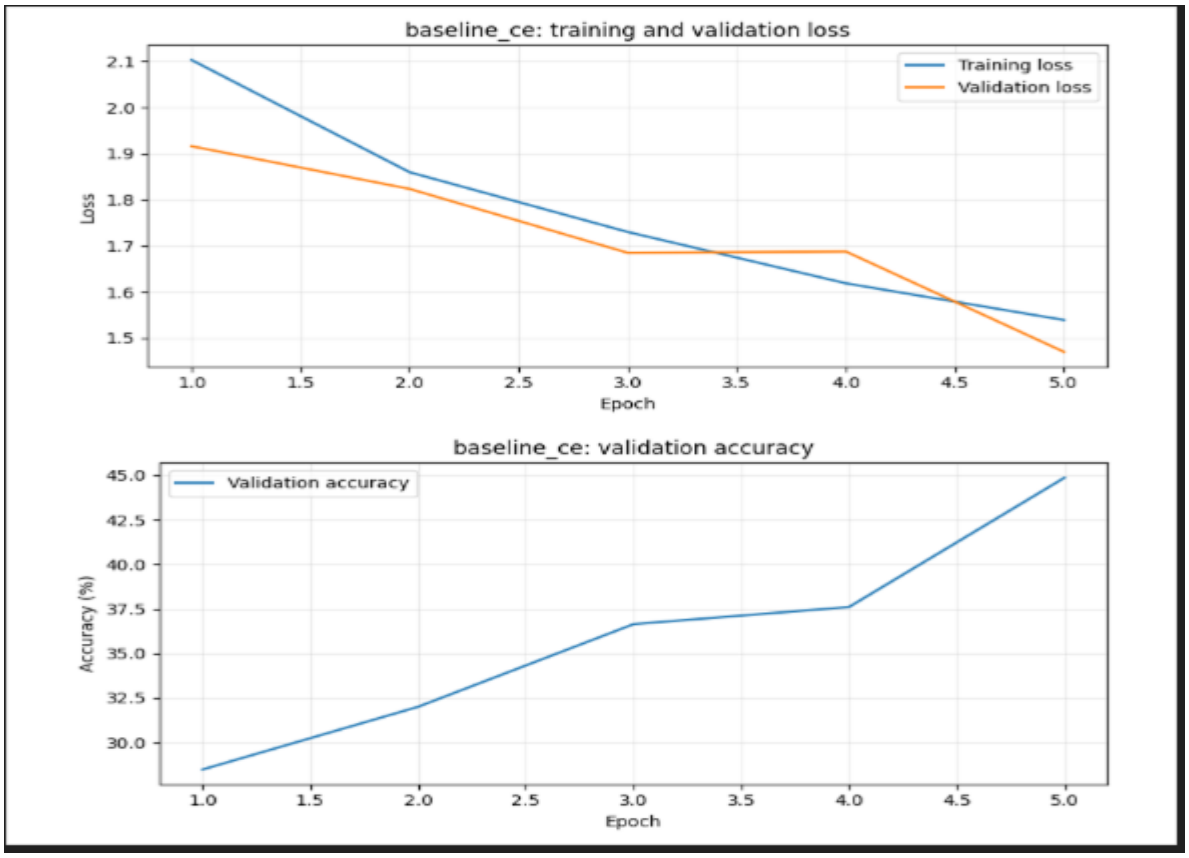
## 5. Results

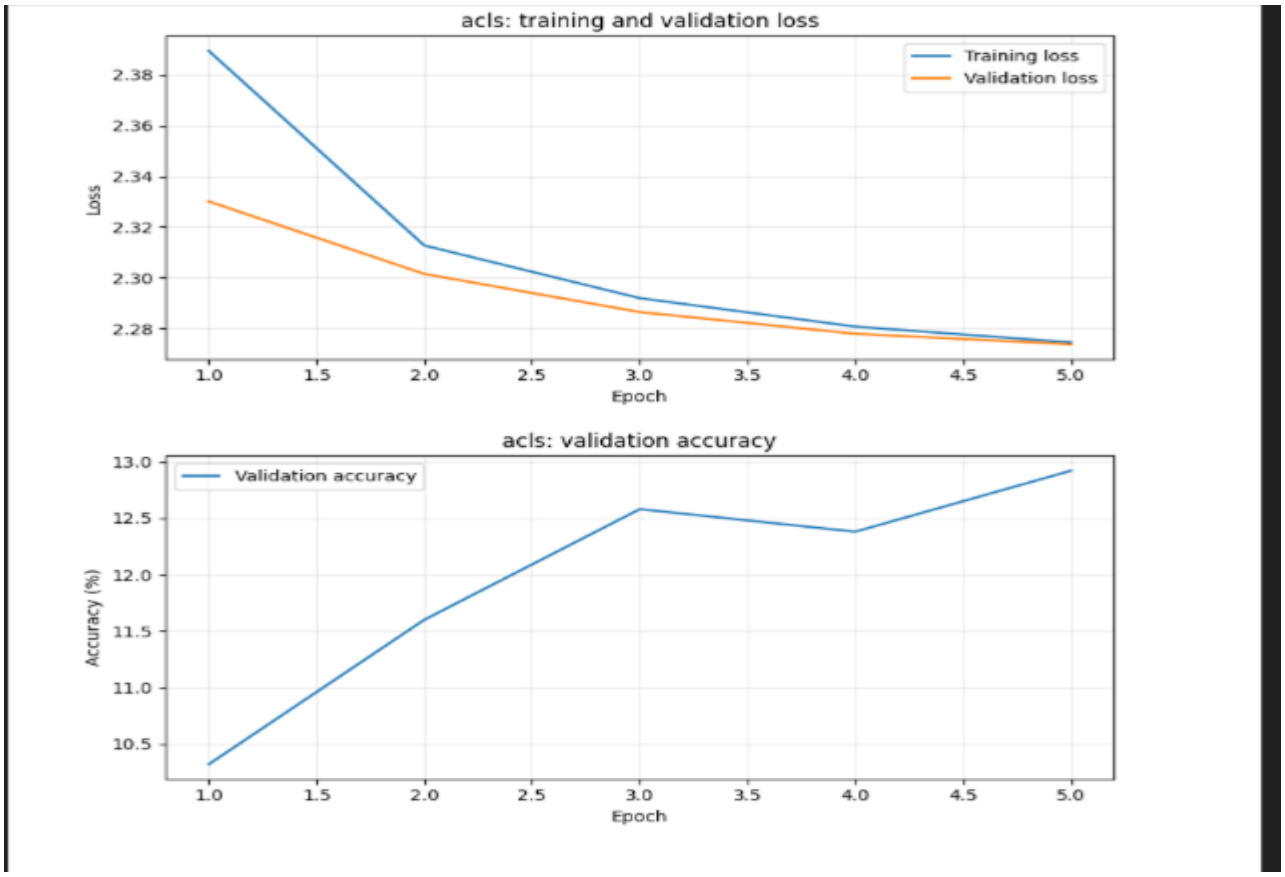
| Model | Accuracy (%) | ECE (%) | AECE (%) |
|---|---|---|---|
| Baseline CE | 45.840001 | 2.387858 | 2.387860 |
| ACLS | 13.249999 | 1.138310 | 1.017359 |

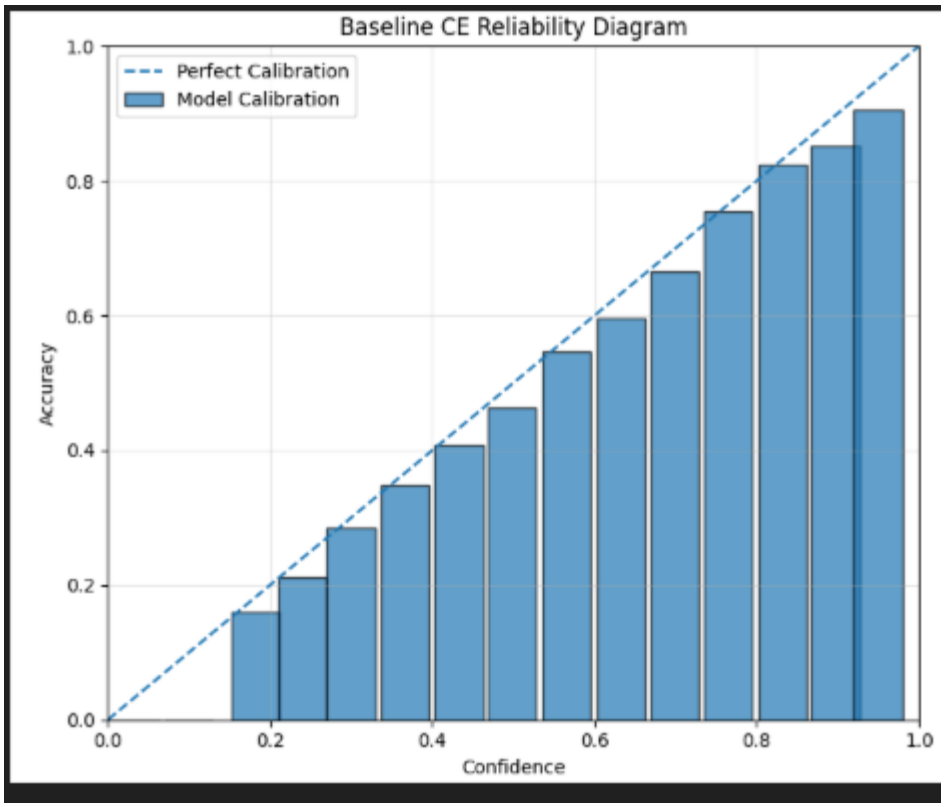
The original paper's CIFAR-10 reference values reported in the selected paper are shown below. The paper reports validation-split results; this reproduction reports results on a held-out test split. They are therefore reference values rather than a strictly like-for-like comparison.

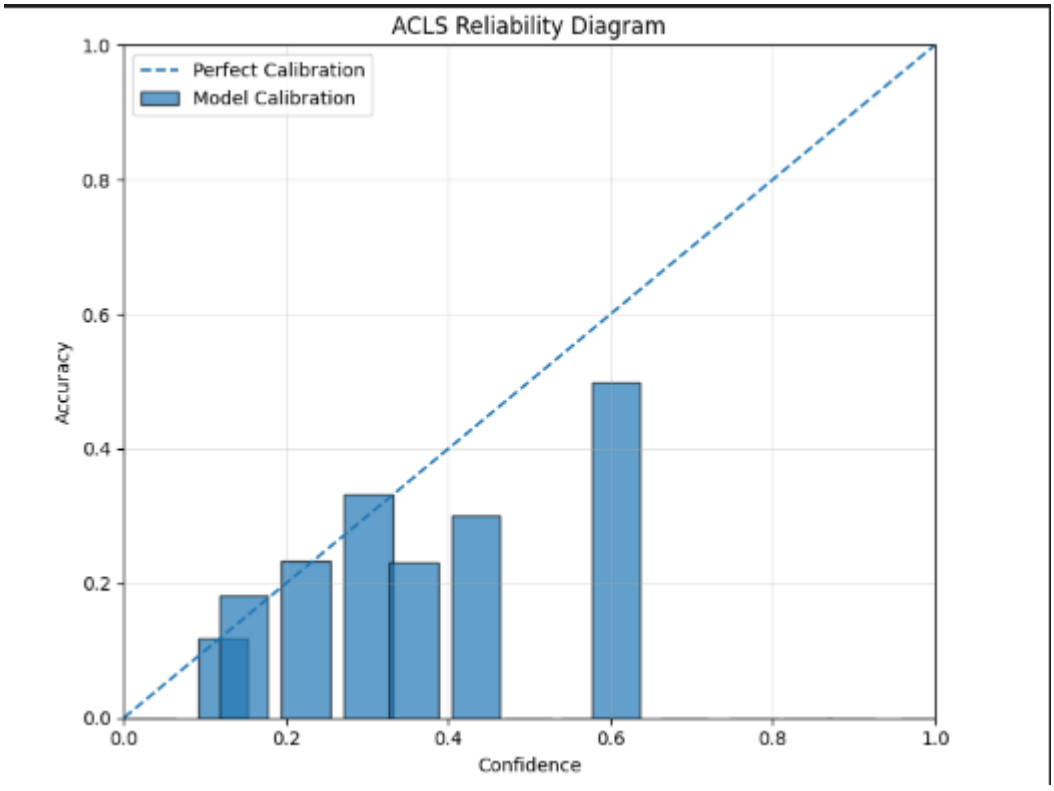
| Model | Original accuracy (%) | Original ECE (%) | Original AECE (%) |
|---|---|---|---|
| Baseline CE | 93.20 | 5.85 | 5.84 |
| ACLS | 95.40 | 1.12 | 2.87 |

The reproduced results show that the Baseline CE model achieved higher classification accuracy than the ACLS model. However, ACLS obtained lower ECE and AECE values in this experiment. This indicates that calibration error and classification accuracy measure different aspects of model performance. The results also show that the ACLS implementation did not achieve the expected classification performance under the current experimental settings.


baseline_ce: training and validation loss
Training loss
Validation loss
Loss
Epoch
baseline_ce: validation accuracy
Validation accuracy
Accuracy (%)
Epoch


acls: training and validation loss
Training loss
Validation loss
Loss
Epoch
acls: validation accuracy
Validation accuracy
Accuracy (%)
Epoch


Baseline CE Reliability Diagram
Perfect Calibration
Model Calibration
Accuracy
Confidence
0.0
0.2
0.4
0.6
0.8
1.0


ACLS Reliability Diagram
Perfect Calibration
Model Calibration
Accuracy
Confidence
0.0
0.2
0.4
0.6
0.8
1.0

## 6. Discussion and Reproducibility Challenges

The reproduction provides an implementation-level comparison between a CE baseline and an ACLS-based model, but it does not claim exact reproduction of the paper. The most important implementation challenge was instability during ACLS optimization. Very large gradient values and non-finite loss were observed in an earlier run. The later experiment used a smaller learning rate and gradient clipping to reduce this problem.

These stability adjustments are important when interpreting the results. A change in learning rate or gradient handling can affect both the learning process and final model performance. In addition, the experiment used a limited five-epoch training schedule, whereas the paper's reported CIFAR-10 setup used a much longer schedule. Differences in training duration, optimizer settings, architecture adaptation, data split, and software environment can all contribute to differences in accuracy and calibration.

The original paper's reported figures are based on its validation protocol, while this notebook calculates metrics on the held-out test set. As a result, differences between the two sets of numbers cannot be attributed to the ACLS method alone. The comparison is useful as a documented reproduction attempt, but it is not a controlled reproduction under identical conditions.

A further limitation is that the current implementation and experiment budget may not reproduce every detail of the authors' training pipeline. The exact settings used for the final runs should be checked against experiment_config.json and the notebook before the report is submitted.

## 7. Conclusion and Next Steps

This assignment implemented a CIFAR-10 image-classification pipeline using an adapted ResNet-50 architecture and compared a Cross-Entropy baseline with an ACLS-based objective. The work includes dataset preprocessing, model implementation, training, test evaluation using Accuracy/ECE/AECE, reliability diagrams, and a comparison with reference values from the original paper.

The experiment also demonstrated practical reproducibility challenges, particularly ACLS optimization instability and differences between the assignment's experimental setup and the original paper. The results are reported as obtained, and the limitations are documented rather than presenting the work as an exact replication. Future work could use a longer training schedule, verify the loss implementation against the paper and official code, and align the evaluation protocol more closely with the original study.

## References

[1] H. Park, J. Noh, Y. Oh, D. Baek, and B. Ham, “ACLS: Adaptive and Conditional Label Smoothing for Network Calibration,” *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.

[2] J. Ni, H. Zhao, J. Gao, D. Guo, and H. Zha, “Balancing Two Classifiers via A Simplex ETF Structure for Model Calibration,” *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025, pp. 30712–30721.

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[5] PyTorch Documentation, https://pytorch.org/docs/

[6] Torchvision Documentation, https://pytorch.org/vision/stable/

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